AN AI-DRIVEN CYBERSECURITY FRAMEWORK WITH BLOCKCHAIN-BASED AUDITABILITY FOR SECURE THREAT INTELLIGENCE

Enterprise-grade security management powered by distributed blockchain ledgers, federated intelligence, and proactive AI-driven monitoring.

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INTRODUCTION

- **Intelligent Real-Time Orchestration**: Our project delivers an enterprise-grade Cybersecurity Portal powered by intelligent agents that manage servers and firewalls in real time.
- **Decentralized Intelligence**: These agents enforce configurations, collect logs, and run local anomaly detection using advanced Machine Learning (ML) algorithms within a Federated Learning (FL) framework.
- **Breach Detection & AI Analysis:** The system proactively identifies security compromises by detecting faulty logs and anomalous data patterns, automatically generating AI-driven analysis reports with quantified risk scoring to provide deep visibility into threat severity.
- **Blockchain-Backed Integrity:** A permissioned Blockchain provides immutable audit trails and transparent policy governance, ensuring verifiable trust and automated rollbacks across teams.

WHY IT MATTERS

1. **Rising Cyber Threats**
   Modern enterprises face increasingly sophisticated attacks targeting servers, firewalls, and cloud infrastructure. Traditional monitoring tools are often reactive rather than proactive.

2. **Compliance & Accountability**
   Organizations must maintain verifiable logs and approval workflows to comply with security standards and regulatory frameworks. Lack of tamper-proof records can lead to legal and financial consequences.

3. **Secure Collaboration**
   Threat intelligence sharing is essential, but exposing raw logs violates privacy and compliance norms. A privacy-preserving collaborative approach is critical.

4. **Faster Incident Response**
   Manual log correlation and analysis delay containment. Automated, data-driven breach intelligence significantly reduces response time.

5. **Need for Quantified Risk**
   Security decisions require measurable risk scoring and severity classification rather than isolated alerts, enabling informed and prioritized action.

PROBLEM STATEMENT

Enterprises struggle to achieve secure, scalable, and collaborative management of servers and firewalls in modern distributed environments. Current systems suffer from:

Fragmented & Non-Transparent Security

Traditional systems lack tamper-evident audit trails and centralized governance, reducing trust and accountability.

No Privacy-Preserving Collaboration

Organizations cannot perform cross-site anomaly detection without sharing sensitive raw logs.

Slow & Manual Incident Response

Breach analysis requires manual log correlation, with no automated AI-driven detection or reporting.

Limited Scalability & Compliance

Existing solutions struggle with cloud-native scalability, multi-stakeholder approvals, and evolving compliance needs.

LITERATURE SURVEY

Shift to Decentralized Security Governance

Recent research highlights the limitations of centralized SIEM (Security Information and Event Management) systems, advocating for Blockchain-based immutable ledgers to prevent log tampering and ensure transparent administrative auditing.

Privacy-Preserving Threat Detection

Literature on Federated Learning (FL) demonstrates that security models can be trained across distributed nodes (agents), allowing for global threat intelligence without exposing sensitive local network data.

Anomaly Detection in High-Dimensional Logs

Studies in Unsupervised Machine Learning prove highly effective at identifying "sparse" malicious patterns—such as the faulty or corrupted logs used in modern obfuscation attacks.

Evolution of Autonomous Response

Modern cybersecurity frameworks are moving toward Autonomous AI Agents that utilize Rule-Based Engines to reduce "Mean Time to Respond" (MTTR), enabling real-time actions like automated policy rollbacks during active breaches.

Quantified Risk & Explainable AI

Current trends emphasize the need for Quantified Risk Scoring rather than binary alerts, providing security teams with AI-driven analysis reports that explain the "severity" and "intent" of detected compromises.

OUR OBJECTIVES

Centralized Security Management

Develop a unified portal for managing server and firewall policies with approval workflows and rollback capabilities.

Immutable Audit & Governance

Ensure tamper-evident logging of all administrative actions using blockchain-based smart contracts.

Privacy-Preserving Anomaly Detection

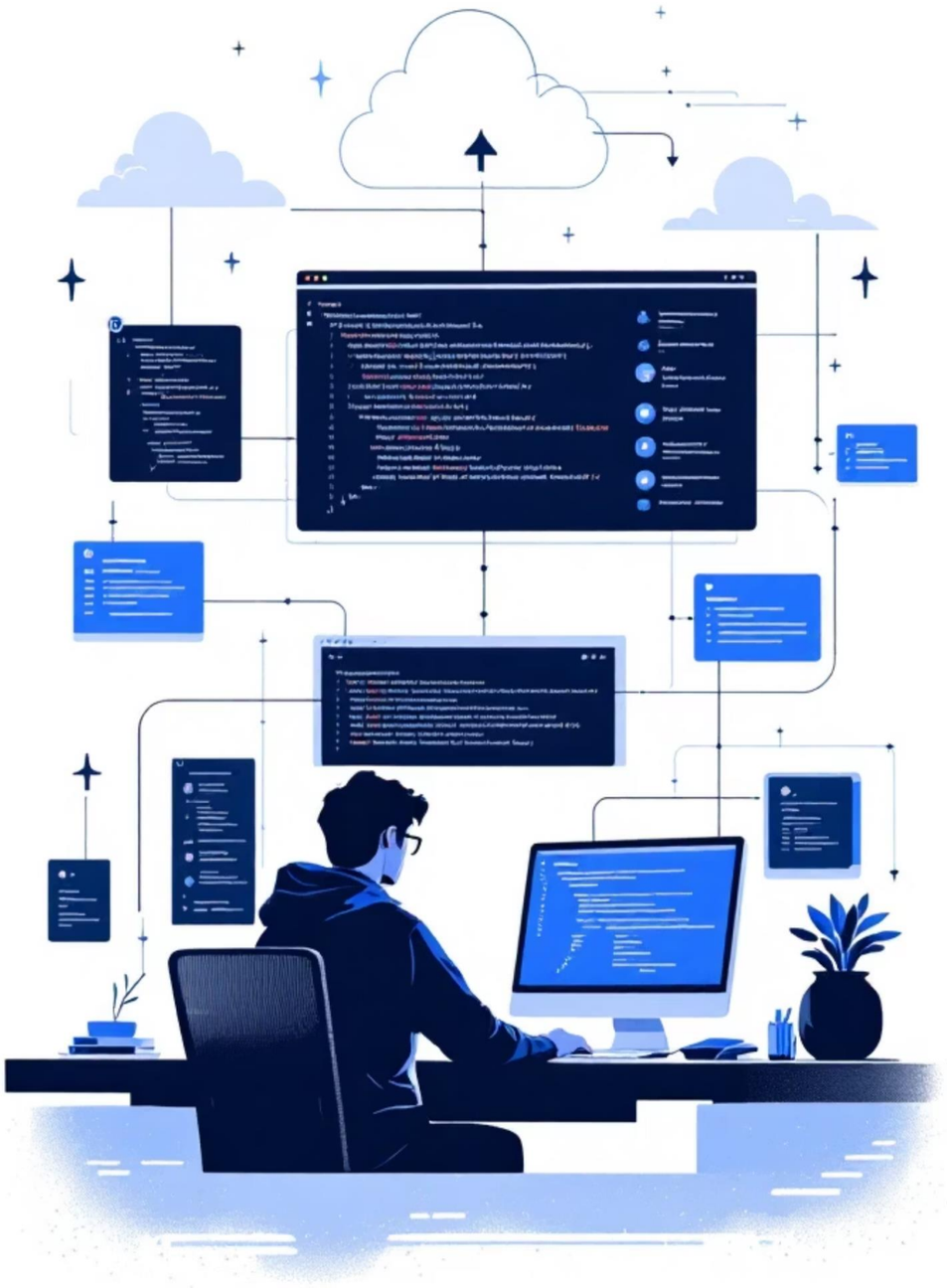
Enable collaborative threat detection using Federated Learning without sharing raw log data.

Intelligent Breach Analysis

Automate breach detection through AI-driven risk reports, triggering immediate policy rollbacks and system freezes to neutralize threats in real time.

Innovation & scaling

Implement a secure, multi-tenant architecture with role-based access control and real-time monitoring.



IMPLEMENTATION

Phase 1: Foundation

Architecting the Vite + React portal and Node.js/Postgres backend to establish a secure, role-based management dashboard for security operations.

Phase 2: Integration

Deploying intelligent agents for real-time monitoring and integrating Hyperledger Fabric to ensure an immutable blockchain audit trail for all system logs.

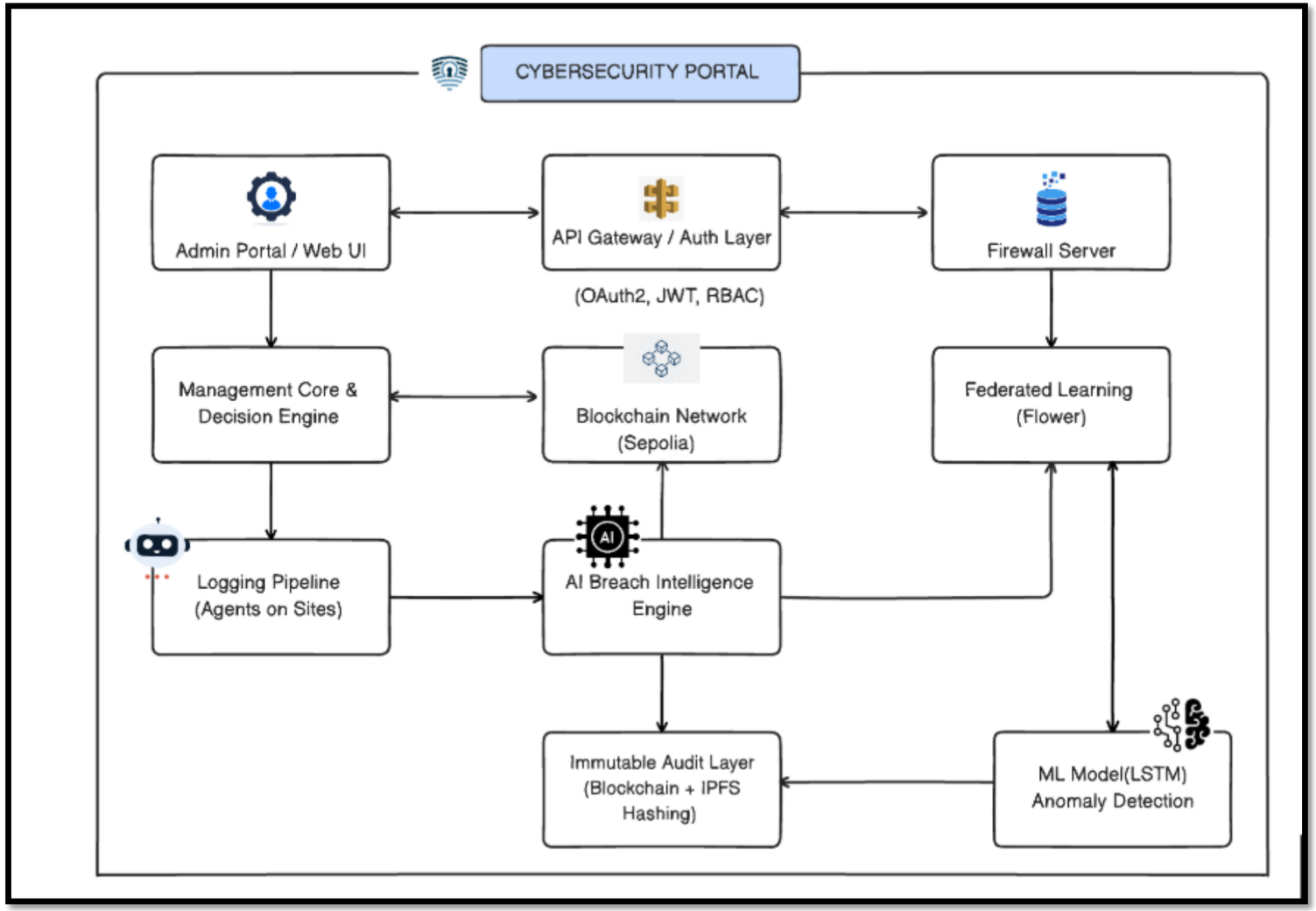
Phase 3: Intelligence

Implementing Federated Learning to detect high-risk anomalies, while enabling autonomous agents to trigger immediate policy rollbacks and system freezes during breaches.

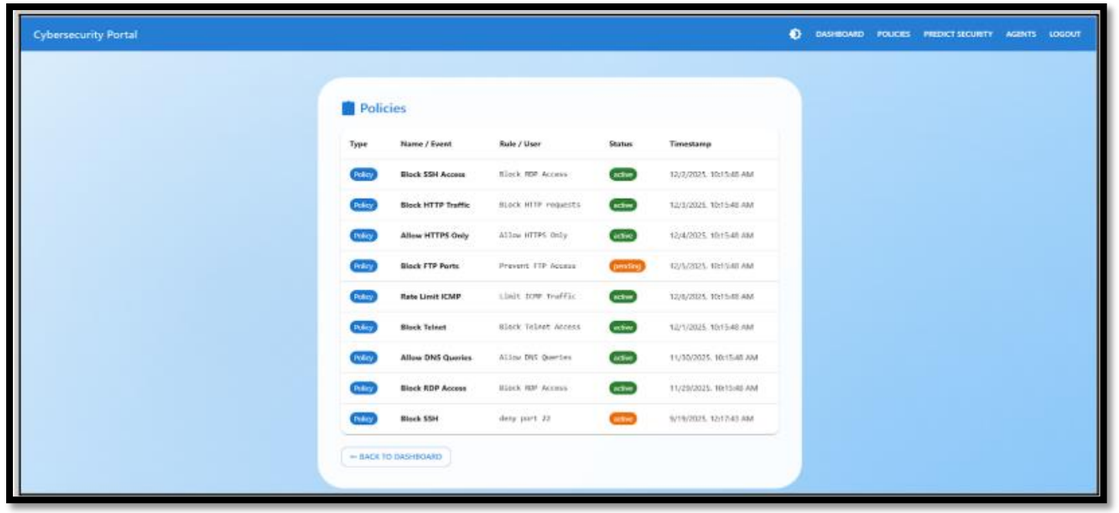
Phase 4: Deployment

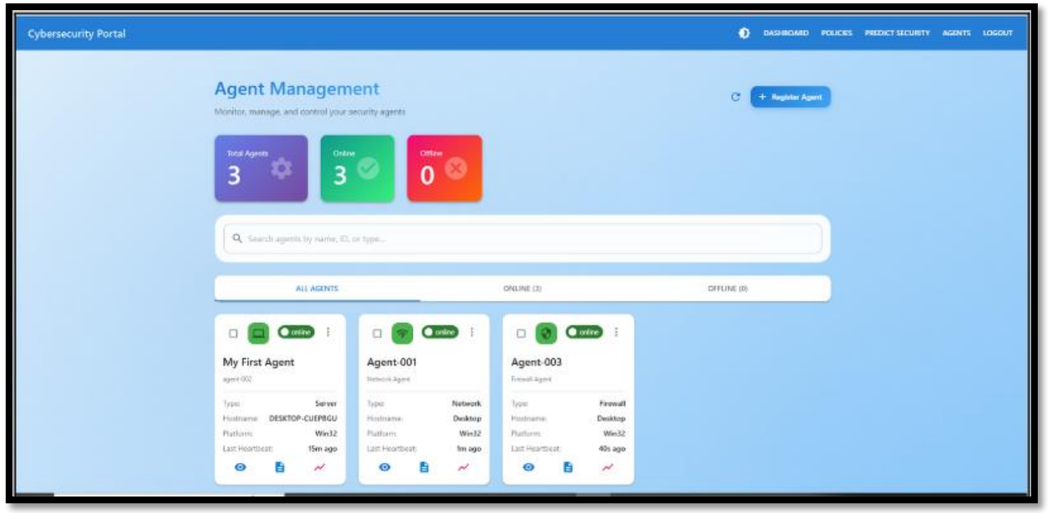
Transitioning to a cloud-native environment using Kubernetes and Terraform, followed by rigorous attack simulation to verify the platform's automated response capabilities.

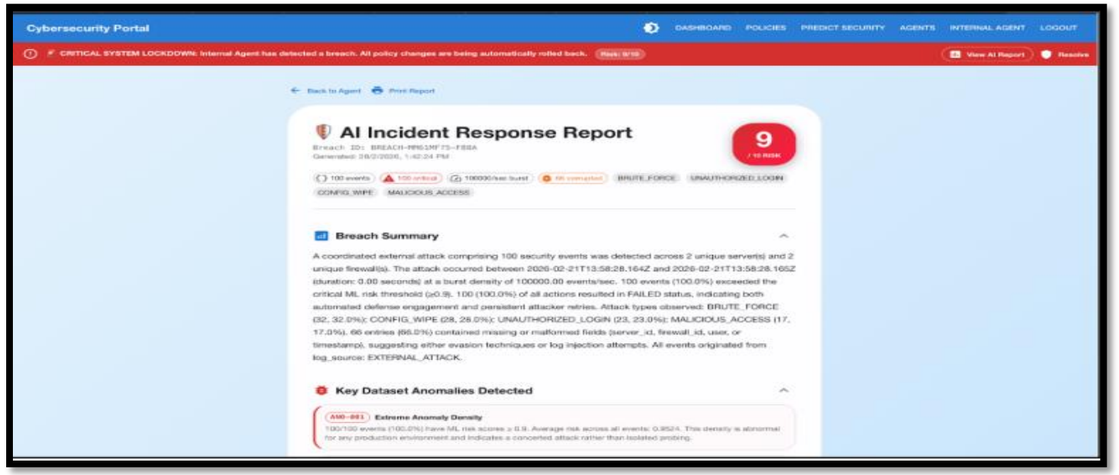
SYSTEM ARCHITECTURE

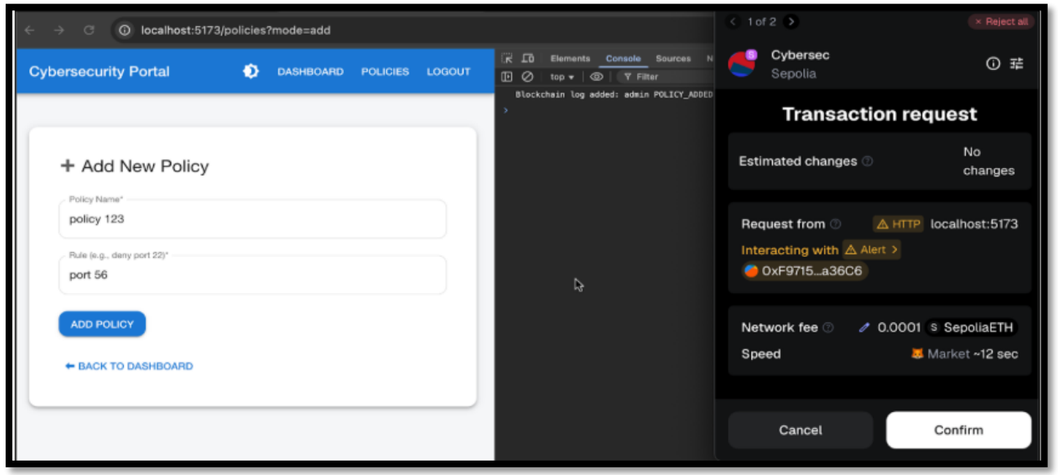


PROTOTYPE


Cybersecurity Portal
DASHBOARD
POLICIES
PREDICT SECURITY
AGENTS
LOGOUT
Policies
Type
Name / Event
Rule / User
Status
Timestamp
Block SSH Access
Block HTTP Traffic
Allow HTTPS Only
Block FTP Ports
Rate Limit ICMP
Block Telnet
Allow DNS Queries
Block RDP Access
Block SSH
BACK TO DASHBOARD


Cybersecurity Portal
DASHBOARD
POLICIES
PREDICT SECURITY
AGENTS
LOGOUT
Agent Management
Monitor, manage, and control your security agents
Register Agent
Total Agents
3
Online
3
Offline
0
ALL AGENTS
ONLINE (3)
OFFLINE (0)
My First Agent
Agent-001
Agent-003


Cybersecurity Portal
DASHBOARD
POLICIES
PREDICT SECURITY
AGENTS
INTERNAL AGENT
LOGOUT
CRITICAL SYSTEM LOCKDOWN: Internal Agent has detected a breach. All policy changes are being automatically rolled back.
View AI Report
Resolve
Back to Agent
Print Report
AI Incident Response Report
9
Breach Summary
Key Dataset Anomalies Detected


localhost:5173/policies?mode=add
Cybersecurity Portal
DASHBOARD
POLICIES
LOGOUT
Add New Policy
Policy Name*
policy 123
port 56
ADD POLICY
BACK TO DASHBOARD
Elements
Console
Sources
Blockchain log added: admin POLICY_ADDED
1 of 2
Reject all
Cybersec
Sepolia
Transaction request
Estimated changes
No changes
Request from
HTTP
localhost:5173
Interacting with
Alert
0xF9715...a36C6
Network fee
0.0001
SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm

NOVEL CONTRIBUTION

1

Unified Integration Architecture

First integrated platform combining blockchain-based governance, federated anomaly detection, intelligent agents, and AI-driven breach intelligence within a single enterprise cybersecurity system.

2

Tamper-Evident Policy Governance

Implementation of smart contract-based approval workflows ensuring immutable logging and transparent multi-stakeholder security management.

3

Privacy-Preserving Collaborative Detection

Federated anomaly detection framework enabling collaborative threat identification across distributed systems while preserving data privacy through decentralized model learning.

4

Deterministic AI Breach Analysis Engine

AI-driven breach analysis engine that detects anomalous activities, automatically generating breach reports with quantified risk assessment and enabling policy rollback for rapid threat mitigation.

FUTURE VISION

1

Advanced Intelligent Threat Detection

Incorporate adaptive AI models and contextual behavioral analysis to improve anomaly detection accuracy and reduce false positives.

2

Scalable Enterprise Deployment

Transition towards containerized and cloud-native architecture supporting high availability, distributed processing, and multi-organization environments.

3

Optimized Blockchain-Based Governance

Enhance smart contract efficiency through optimized consensus mechanisms for faster and cost-effective tamper-evident audit management.

4

Privacy-Enhanced Collaborative Learning

Strengthen federated learning frameworks using secure aggregation and privacy-preserving computation for cross-domain threat intelligence sharing.

5

Autonomous Security Response

Enable intelligent agent-driven automated response workflows for real-time policy enforcement, incident containment, and recovery orchestration.

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THANK YOU

Open for Questions & Discussion